

Smart Agriculture: Communication Explanation

This document explains how communication works in the Smart Agriculture Project, an IoT-based irrigation system. The system uses an ESP8266 NodeMCU to connect sensors, a water pump, and online platforms (ThingSpeak and Blynk) to control irrigation based on soil moisture, weather, time, and user input. Below, we describe each communication "axe" (pathway) in simple terms.

1. Overview

The system communicates to:

- **Read data:** Soil moisture sensor, weather forecasts (ThingSpeak), and time (NTP servers).
- **Control devices:** Turn the water pump and LEDs on/off.
- **Share data:** Send soil moisture to ThingSpeak and display status on Blynk.
- **Accept user input:** Manual pump control via Blynk.

All communication is managed by the ESP8266, which uses Wi-Fi to connect to the internet. The MATLAB script runs separately to fetch weather data and update ThingSpeak.

2. Communication Axes

Each "axe" represents a specific data flow between components. Below, we explain each one simply.

Axe 1: Soil Moisture Sensor → ESP8266

- **What happens:** The soil moisture sensor measures how wet or dry the soil is and sends a number (0–1023) to the ESP8266.
- **How it works:**
 - The sensor is connected to the ESP8266's analog pin **A0**.
 - The ESP8266 reads the sensor value using `analogRead(soilPin)`.
 - A value > 600 means the soil is dry, triggering further checks.
- **Purpose:** Tells the system if the soil needs water.
- **Frequency:** Every 30 seconds (unless manual mode is active).

Axe 2: ESP8266 → Water Pump (via Relay)

- **What happens:** The ESP8266 turns the water pump on or off based on soil, weather, and time.
- **How it works:**
 - The pump is connected to a **5V relay** (signal pin to ESP8266's **D2**).

- The ESP8266 sends a HIGH signal to `pumpRelayPin` to turn the pump on, or LOW to turn it off.
 - The relay uses a separate 4.5V power source to operate the pump safely.
- **Purpose:** Waters the crops when conditions are right (dry soil, no rain, correct time).
- **Frequency:** When needed (automatic mode) or for 10 seconds (manual mode).

Axe 3: ESP8266 → LEDs

- **What happens:** The ESP8266 lights up LEDs to show system status.
- **How it works:**
 - Three LEDs are connected to ESP8266 pins:
 - **Pump LED (D1):** Lights up when the pump is on.
 - **Rain LED (D6):** Lights up if rain is expected or it's not the right time to water.
 - **Status LED (D7):** Not actively used in the code but defined for general status.
 - The ESP8266 sends HIGH or LOW signals to these pins using `digitalWrite`.
- **Purpose:** Gives visual feedback on what the system is doing.
- **Frequency:** Updates with every decision (e.g., pump on/off, rain detected).

Axe 4: ESP8266 ↔ ThingSpeak

- **What happens:** The ESP8266 sends soil moisture data to ThingSpeak and reads weather data (rain forecast).
- **How it works:**
 - **Sending soil data:**
 - The ESP8266 sends the sensor value to ThingSpeak's field 1 using an HTTP GET request (`sendSoilToThingSpeak`).
 - Example URL: `http://api.thingspeak.com/update?api_key=2OJMR1D255GTCJ24&field1={value}`.
 - **Reading rain forecast:**
 - The ESP8266 fetches the rain forecast (0 or 1) from ThingSpeak's field 3 using an HTTP GET request (`getRainForecast`).
 - Example URL: `http://api.thingspeak.com/channels/2906874/fields/3/last.txt?api_key=I0XZ87IDI1STJ9M0`.
 - Uses Wi-Fi and the `ESP8266HTTPClient` library.
- **Purpose:** Logs soil data for monitoring and checks if rain is coming to avoid unnecessary watering.
- **Frequency:** Every 30 seconds for soil data; when soil is dry for rain data.

Axe 5: ESP8266 ↔ Blynk

- **What happens:** The ESP8266 sends status messages to the Blynk app and receives manual pump commands.
- **How it works:**
 - **Sending status:**
 - The ESP8266 sends logs (e.g., soil value, pump status) to Blynk's virtual pin **V10** using `Blynk.virtualWrite`.
 - **Receiving commands:**
 - The Blynk app sends a 1 or 0 to virtual pin **V0** when the user presses a button.

- The `BLYNK_WRITE(V0)` function triggers: 1 turns the pump on for 10 seconds, 0 turns it off.
 - Uses Wi-Fi and the `BlynkSimpleEsp8266` library.
- **Purpose:** Lets users monitor the system and manually control the pump from their phone.
- **Frequency:** Real-time for status updates; on-demand for manual control.

Axe 6: ESP8266 ↔ NTP Servers

- **What happens:** The ESP8266 gets the current time to check if it's the right time to water (5–9 AM or 6–8 PM).
- **How it works:**
 - Connects to NTP servers (`pool.ntp.org`, `time.nist.gov`) using `configTime`.
 - Retrieves the current time with `time(nullptr)` and extracts the hour using `localtime`.
 - Uses Wi-Fi for internet access.
- **Purpose:** Ensures irrigation happens only during optimal time windows.
- **Frequency:** Checked every 30 seconds when soil is dry.

Axe 7: MATLAB Script ↔ OpenWeatherMap

- **What happens:** A separate MATLAB script fetches weather forecasts and sends rain predictions to ThingSpeak.
- **How it works:**
 - Runs on a computer, not the ESP8266.
 - Sends an HTTP request to OpenWeatherMap's API for 3-day forecasts (72 hours, 3-hour blocks).
 - Example URL: `https://api.openweathermap.org/data/2.5/forecast?lat=36.8&lon=10.17&appid=845a9c746afdf739eaac68ec6e1bb979&units=metric`.
 - Checks for rain (≥ 0.1 mm) in any block; sets `rainExpected` to 1 if found, else 0.
 - Sends `rainExpected` to ThingSpeak's field 3 using `thingSpeakWrite`.
- **Purpose:** Provides the ESP8266 with rain forecast data via ThingSpeak.
- **Frequency:** Not specified, but typically run periodically (e.g., every few hours).

3. How It All Works Together

1. **Sensor Input:** The ESP8266 reads soil moisture (Axe 1).
2. **Decision-Making:**
 - If the soil is dry (>600), it checks the rain forecast (Axe 4) and current time (Axe 6).
 - If no rain and it's 5–9 AM or 6–8 PM, it turns on the pump (Axe 2) and pump LED (Axe 3).
 - If rain is expected or it's not the right time, it skips watering and lights the rain LED.
3. **Monitoring:** Soil data is sent to ThingSpeak (Axe 4), and status updates go to Blynk (Axe 5).
4. **Manual Control:** The user can override via Blynk (Axe 5) to turn the pump on for 10 seconds.
5. **Weather Updates:** The MATLAB script periodically updates ThingSpeak with rain predictions (Axe 7).

4. Key Points

- **Wi-Fi:** The ESP8266 uses Wi-Fi to connect to ThingSpeak, Blynk, and NTP servers.
- **Real-Time:** Blynk provides instant control and monitoring; ThingSpeak logs data for later viewing.

- **Simple Feedback:** LEDs show what's happening locally; Blynk shows it remotely.
- **Weather Integration:** The MATLAB script ensures the system knows about upcoming rain.

5. Limitations

- **Wi-Fi Dependency:** No communication if Wi-Fi fails.
- **MATLAB Dependency:** Weather updates require a separate computer running the script.
- **Fixed Timing:** Irrigation windows are hardcoded; could be adjustable.

6. Conclusion

The system's communication is simple and effective, using the ESP8266 to connect local devices (sensor, pump, LEDs) with online platforms (ThingSpeak, Blynk, NTP). The MATLAB script adds weather data, making the system smart and water-efficient. Each axe ensures data flows smoothly to automate irrigation and keep users informed.